

Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

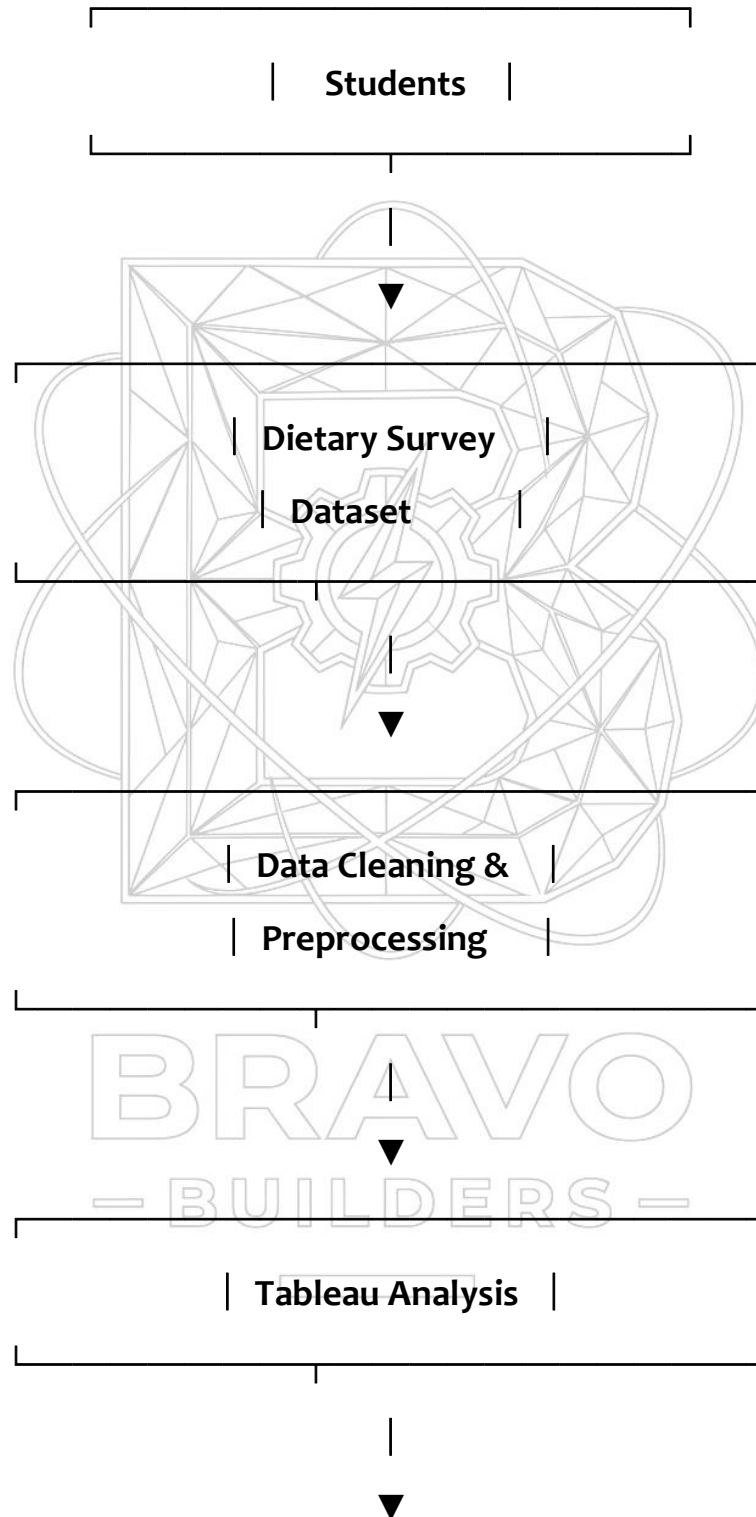
TEAM ID	SWTID-2026-6798
PROJECT TITLE	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
MAX MARKS	3 MARKS

3.3 – Data Flow Diagram

The Data Flow Diagram (DFD) for the project "**Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study**" describes how data moves through the system from collection to visualization and decision-making. The process begins with college students, who act as the primary data source by providing information through dietary surveys. The collected data includes age, gender, meal frequency, daily calorie intake, favorite cuisine, vegetable consumption, dining-out frequency, exercise habits, income level, healthy eating perception, and overall health rating. This information is stored in a centralized dataset where it undergoes data cleaning, validation, and preprocessing to remove errors, duplicates, and missing values. After preprocessing, the refined dataset is imported into Tableau, where the data is analyzed and transformed into meaningful visualizations such as bar charts, pie charts, scatter plots, box plots, dashboards, and stories. Tableau processes the information to identify dietary trends, food preferences, nutritional gaps, and relationships between eating habits, exercise, and health outcomes. The generated dashboards allow users to interact with filters and compare data across different demographic groups, enabling deeper analysis. The insights produced by the system are then shared with students, nutritionists, educators, and college administrators, helping them understand current dietary patterns and make informed decisions. Based on these findings, practical dietary strategies, health awareness programs, and healthier campus food policies can be developed. Thus, the data flows continuously from students → survey data → data preprocessing → Tableau analysis → interactive dashboards → insights and recommendations → stakeholders, ensuring that

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raw data is converted into actionable knowledge for improving students' health and well-being.



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